An EBS volume is guaranteed to be available for 99.999% of the time. So if an availability zone outage occurs, the volume is no longer available. This does not imply that you lose any data. As soon as the availability zone is back online, you can access the EBS volume again with all its data.

An EBS volume guarantees that you won't lose any data in 99.9% of the time. This guarantee is called the durability of the EBS volume. If you have 1,000 volumes in use, you can expect that you will lose one of the volumes and its data a year.

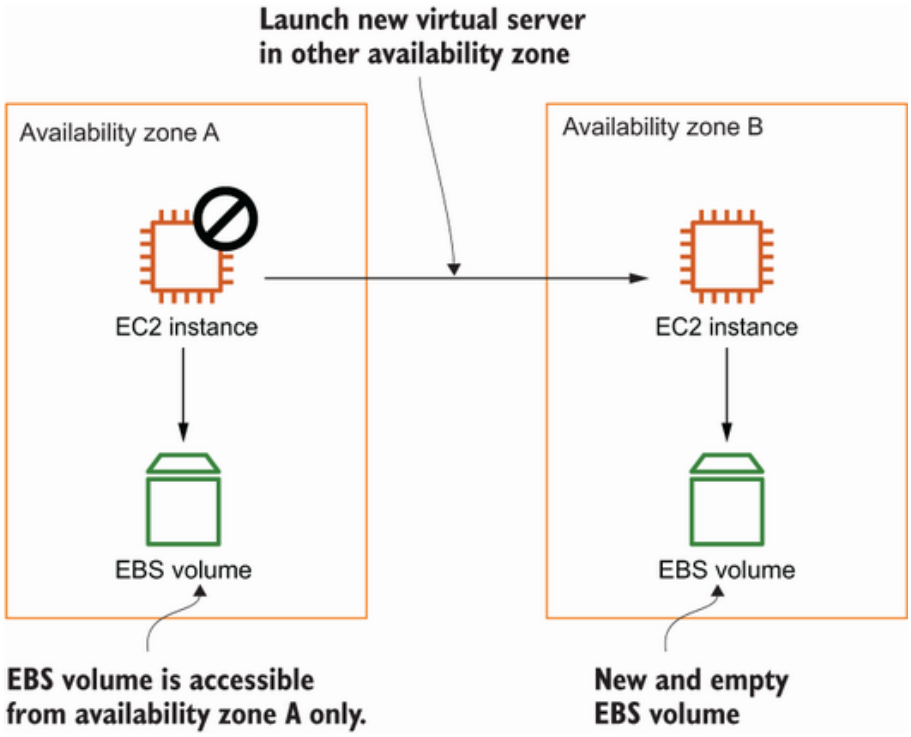


Figure 13.5 An EBS volume is available only in a single availability zone.

You have multiple solutions for this problem:

- Outsource the state of your virtual machine to a managed service that uses multiple availability zones by default: RDS, DynamoDB (NoSQL database), EFS (NFSv4.1 share), or S3 (object store).
- Create snapshots of your EBS volumes regularly, and use these snapshots if an EC2 instance needs to recover in another availability zone. EBS snapshots are stored on S3 and, thus, are available in multiple availability zones. If the EBS volume is the root volume of the ECS instance, create AMIs to back up the EBS volume instead of a snapshot.